

# HEAP 1

Develop Max heap which contains the following functions

- Add data to this Max heap
- Remove data from this Max heap
- Show all data in this Max heap

Example

| Input | Result   | Description                    |
|-------|----------|--------------------------------|
| a 12  |          | add 12 to this Max heap        |
| a 13  |          | add 13 to this Max heap        |
| a 20  |          | add 20 to this Max heap        |
| p     | 20 12 13 | Show all data in this Max heap |
| d     | 20       | Show a removed data            |
| p     | 13 12    | Show all data in this Max heap |
| e     |          | End                            |

| Input | Output                               |
|-------|--------------------------------------|
| a 18  | 19 18 2 10                           |
| a 19  | 258 19 95 10 18 2                    |
| a 2   | 258 98 102 36 18 2 95 10 19          |
| a 10  | 258 98 102 36 18 2 95 10 19          |
| p     | 258 98 102 36 71 2 95 10 19 18       |
| a 258 | 258 98 102 36 71 2 95 10 19 18 11    |
| a 95  | 258 98 102 36 71 71 95 10 19 18 11 2 |
| p     | 258                                  |
| a 102 | 102                                  |
| a 36  | 150 98 95 36 71 71 2 10 19 11 18     |
| a 98  | 150                                  |
| p     | 98                                   |
| p     | 95                                   |
| a 71  | 71                                   |

|        |                     |
|--------|---------------------|
| p      | 71 36 11 19 18 10 2 |
| a 11   |                     |
| p      |                     |
| a 71   |                     |
| p      |                     |
| d      |                     |
| d      |                     |
| a 150  |                     |
| p      |                     |
| d      |                     |
| d      |                     |
| d      |                     |
| d      |                     |
| p      |                     |
| s 2521 |                     |
| s 2158 |                     |
| s 3010 |                     |
| s 3629 |                     |
| s 1234 |                     |
| e      |                     |